PMATH 336 Review

PMATH 336 - Introduction to Group Theory with Applications

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Lecture 3

Real General Linear Group

$$GL_n(\mathbb{R}) := \{A \in M_{n\times n}(\mathbb{R}) : \text{A is invertible } (\det(\text{A}) \in \mathbb{R}^*)\}$$

- Non-abelian

Note: $\mathbb{Z}^* = \{-1, 1\}$

Group Axioms

1. For any elements $a, b, c \in G$, $(a\cdot b)\cdot c = a\cdot (b\cdot c)$ (this says that $\cdot$ is **associative**).
2. There exists an element $e \in G$ (**identity, unique**), such that $a\cdot e = e\cdot a = a$ for all $a \in G$.
3. For all $a \in G$, there exists an element $a^{-1} \in G$ such that $a\cdot a^{-1} = a^{-1}\cdot a = e$. (**inverse, unique**)

Proposition 3: (Cancellation Property)

Let $a, b, c \in G$ for a group G. If $ab = ac$, then $b = c$. If $ba = ca$, then $b = c$.

Lecture 4

Dihedral Group D_n

symmetry: if you don't look at it, you won't know it happens.

D_n $(n \geq 3)$: non-abelian

$|D_n| = 2n$ (n rotations + n flips)

Symmetric Group

S_n $(n \geq 3)$: non-abelian

$|S_n| = n!$

Direct Products

Let $(G,\ *)$ and $(H,\ \circ)$ be groups.

$(G \times H,\ \cdot\ ) := \{(g, h) : g \in G,\ h \in H\}\ and\ (g_1,\ h_1)\cdot(g_2,\ h_2) := (g_1 * g_2,\ h_1 \circ h_2)$.

Lecture 6

trivial subgroup $:= \{e\}$

Subgroup Test

Let $(G,\cdot\ )$ be a group and $H \subset G$ a non-empty subset of the group. Suppose the following two conditions hold:

1. For all $a, b \in H, a\cdot b \in H$, and

2. For all $a \in H, a^{-1} \in H$.

Then H is a subgroup of G.

Coset

Let G be a group and $H < G$ a subgroup, and $a \in G$. The set $aH := \{ah : h \in H\}$ is called a left coset of H in G. The set $Ha := \{ha : h \in H\}$ is called a right

coset of H in G.

- index of H in G, denoted $|G : H|$, is the number of left cosets of H in G. (if G is finite, then $|G|/|H| = |G : H|$).
- left multiplication by an element is a bijection (can tell it from no repeated element in any row/column of a Cayley Table)
- Suppose aH and bH are two left cosets of H in G. Then either they are equal or disjoint.

  Proof:

  Suppose $aH \cap bH = \emptyset$, then we are done. If not, there exists some element in both, let it be x.

  Then $x = ah_1 = bh_2$ for some elements $h_1, h_2 \in H$. We then have $b^{-1}ah_1 = h_2$. Now show that $aH \subset bH$. Let ah be an arbitrary element of aH.

  Then $ah = bb^{-1}ah_1h_1^{-1}h = bh_2h_1^{-1}h$.

  Since $h_1, h_2, h \in H$, and H is a group, $h_2h_1^{-1}h \in H$. Therefore $ah \in bH$ and we can conclude that $aH \subset bH$.

  A similar argument shows $bH \subset aH$ and we have $aH = bH$.

Intersection of two subgroups

If H and K are two subgroups of G, then $H \cap K < G$.

Corollary 14

Any group of prime order is cyclic.

Cyclic group

A group G is cyclic if there exists an element $a \in G$ such that $\langle a \rangle = G$. Such an element is called a generator of G.

Fermat's Little Theorem

If p is prime, then for any integer a we have $a^p \equiv a \; mod \; p$

Lecture 7

Symmetric Groups

- transposition: 2-cycle
- Every cycle can be written as a product of transpositions.
  - For $n > 2, (a_1 \; a_2 ... \; a_n) = (a_1 \; a_n)(a_1 \; a_2 ... \; a_{n-1})$.
  - $(a_1 \; a_2 ... \; a_n) = (a_1 \; a_n)(a_1 \; a_{n-1}) ... (a_1 \; a_2)$
- Every permutation can be written as a product of disjoint cycles.
- Every permutation can be written as a product of transpositions.
- The identity $(1) \in S_n$ cannot be written as a product of an odd number of transpositions.
- If a permutation $\sigma \in S_n$ can be written as a product of an even number of transpositions, σ is an even permutation. If $\sigma \in S_n$ can be written as a product of an odd number of transpositions, we say σ is an odd permutation.
  - An odd cycle is an even permutation.
- **Alternating group on n letters**: $A_n := \{\sigma \in S_n : \sigma \text{ is even}\}$. A_n is a subgroup of S_n.
  - $|S_n : A_n| = 2$

Lecture 8

Cyclic Groups

- abelian
- Every subgroup of a cyclic group is cyclic.
  - all subgroups of $\mathbb{Z}$ are cyclic.

- Proof:

Let $G = \langle a \rangle$, and let $H < G$. If $H = \{e\}$, then $H = \langle e \rangle$, so assume H is non-trivial.

Let $b \in H$ s.t. $b \neq e$. Then $b = a^s$ for some integer $s \neq 0$.

Let m be the smallest positive integer s.t. $a^m \in H$. Now show $H = \langle a^m \rangle$.

Let $y \in H$ be an arbitrary element, so $y = a^n$ for some n. Let $n = qm + r$ for some integers q, r with $0 \leq r < m$. Then $y = a^{qm+r} = a^{qm}a^r$. So $a^r = y(a^m)^{-q}$.

Since $y, a^m \in H$, $a^r \in H$. But $r < m$, so $r = 0$ and $y = (a^m)^q$. Since every element in H is a power of a^m, $H = \langle a^m \rangle$.

- Let $G = \langle a \rangle$ be a cyclic group of order n. For every positive integer d that divides n, there exists a unique subgroup of order d, the subgroup $H = \langle a^{n/d} \rangle$.
- Let $G = \langle a \rangle$ be a cyclic group of order n. Then for any element $a^s \in G, |a^s| = n/gcd(n, s)$
- Let G be a group of order n, then

  G is cyclic $\iff$ For any positive divisor d of n, G has at most one subgroup of size d.

  $\iff$ For any positive divisor d of n, G has exactly one subgroup of size d.
- Let $G = \langle a \rangle$ with $|G| = n$. Then a^s generates G if and only if $gcd(s, n) = 1$.

Ex:

- $\mathbb{Z}_n = \langle 1 \rangle = \langle n - 1 \rangle$.
  - both have order n
- $\mathbb{Z}_{10}^* = \{1, 3, 7, 9\} = \{3^0, 3^1, 3^2, 3^3\} = \langle 3 \rangle$.
- $\mathbb{Z} = \langle 1 \rangle$.

- $3\mathbb{Z} = \langle 3 \rangle$.
- $\{1, -1, i, -i\} = \langle i \rangle$.
- $\{(1), (123), (132)\} = \langle (123) \rangle < S_3$.
- $\langle a \rangle < G$ is cyclic for any group G and any element $a \in G$.

Lecture 9

Homomorphism

Let $(G, \cdot)$ and $(H, *)$ be groups. A group homomorphism from G to H is a function $\varphi : G \to H$ that satisfies $\varphi(a \cdot b) = \varphi(a) * \varphi(b)$ for all $a, b \in G$.

- $\varphi(e) = e$ and $\varphi(a^{-1}) = \varphi(a)^{-1}$
- $ker(\varphi) := \{g \in G : \varphi(g) = e\}$ (subgroup of G)
- $im(\varphi) := \{h \in H : \text{there exists } g \in G \text{ such that } \varphi(g) = h\}$ (subgroup of H)
- **trivial homomorphism** $\varphi : G \to H$ defined by $\varphi(g) = e$ for all $g \in G$.
  - $im(\varphi) = e$ and $ker(\varphi) = G$
- A homomorphism $\phi : G \to H$ is surjective iff $im(\phi) = H$, ϕ is injective iff $ker(\phi) = \{e\}$.

Isomorphism

A homomorphism $\varphi : G \to H$ such that φ is a bijection ($G \cong H$).

- $\text{Non-abelian group } G \ncong \text{abelian group } H$
- Let $G = \langle a \rangle$ and $H = \langle b \rangle$ be cyclic groups. Then $G \cong H$ if and only if $|G| = |H|$.
  - Let G be a cyclic group. Then if $|G| = n, G \cong \mathbb{Z}_n$. If $|G| = \infty, G \cong \mathbb{Z}$.
  - Let p be a prime and let G and H be groups of order p. Then $G \cong H$.

- Let G and H be groups. A homomorphism $\phi : G \to H$ is an isomorphism if and only if there exists a homomorphism $\varphi : H \to G$ such that $\phi\varphi(h) = h$ for all $h \in H$ and $\varphi\phi(g) = g$ for all $g \in G$. We call such a φ the inverse of ϕ, and denote it $\varphi = \phi^{-1}$.
- Proving bijection: 1. prove surjective/injective 2. prove equal size

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Lecture 10

Normal Subgroup

Let $H < G$ be a subgroup. We say H is a normal subgroup of G if for all $g \in G, ghg^{-1} \in H$ for all $h \in H$. We write this $H \lhd G$.

H is a normal subgroup if and only if $H = ker(\varphi)$ for some homomorphism $\varphi : G \to K$ for some group K.

- If G is abelian, then every subgroup H is normal.
- Suppose $H < G$ is the only subgroup such that $|G : H| = n$ for some n. Then H is normal.
- The centre of a group $Z(G) := \{a \in G : ag = ga \text{ for all } g \in G\}$ is a normal subgroup of G.
- Let H be a normal subgroup of G. If G is finite, $|G/H| = |G|/|H|$.
- Let H be a normal subgroup of G. If G is abelian, then G/H is abelian.
- Let H be a normal subgroup of G. If G is cyclic, then G/H is cyclic
- Suppose $H < G$ is such that $|G : H| = 2$. Then $H \lhd G$.
  - $A_n \lhd S_n$
- Let H be a normal subgroup of G, then $gH = Hg \ \ \forall g \in G$.

Quotient Group $(G/H, \cdot)$

Let G be a group and $H \lhd G$. G/H is the set of all left cosets of H in G, and $(g_1H) \cdot (g_2H) := g_1g_2H$. (We will sometimes write gH as $[g]$, the operation is given by $[g_1] \cdot [g_2] = [g_1g_2]$).

- identity: H or $[e]$
- **quotient map** $\pi : G \to G/H$ by $\pi(g) = gH$.

The Quaternions

- $i^2 = j^2 = k^2 = ijk = -1$
- $Q_8 := \{1, -1, i, -i, j, -j, k, -k\}$
- non-abelian
- all subgroups are normal.

Lecture 11

The First Isomorphism Theorem

Let $\varphi : G \to H$ be a homomorphism. Then $G/ker(\varphi) \cong im(\varphi)$.

- Let K be $ker(\varphi)$, Consider $\phi : G/K \to im(\varphi), [g] \to \varphi(g)$

Let $\varphi : G \to H$ be a homomorphism with G and H finite groups. Then $|im(\varphi)|$ divides both $|G|$ and $|H|$.

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Lemma 45

Let $\varphi : G_1 \to G_2$ be a homomorphism.

If H_1 is a subgroup of G_1, then $\varphi(H_1)$ is a subgroup of G_2.

If H_2 is a subgroup of G_2, then $\varphi^{-1}(H_2)$ is a subgroup of G_1.

The Correspondence Theorem

Let $\varphi : G_1 \to G_2$ be a **surjective** homomorphism, and let $K = ker(\varphi)$. There is a one to one correspondence

$$\begin{aligned} \{\text{Subgroups } H \text{ of } G_1 \text{ such that } K < H < G_1\} \longrightarrow & \quad \{\text{Subgroups } \bar{H} \text{ of } G_2\} \\ H \longmapsto & \quad \varphi(H) \\ \varphi^{-1}(\bar{H}) \longleftarrow & \quad \bar{H} \end{aligned}$$

Furthermore, this bijection has the following properties.

1. For subgroups $K < H_1, H_2 < G_1, H_1 < H_2$ if and only if $\varphi(H_1) < \varphi(H_2)$.

2. For $K < H < G_1, |G_1 : H| = |G_2 : \varphi(H)|$. (here we refer to the ratio regarding the size of G_1 and H, not the number of such subgroups H are in G_1, which the number of subgroups H such that $K < H < G_1$ is the same as the number of subgroups of G_2)

3. A subgroup H containing K is normal in G_1 if and only if $\varphi(H)$ is normal in G_2 .

Corollary 47

Let G and H be groups such that $G \cong H$. Then

1. G and H have the same subgroup lattices.

2. G and H have the same number of subgroups of index k for every k.

Corollary 48

Let G be a group and $K \lhd G$ a normal subgroup. Then there is a one to one correspondence between subgroups of G containing K and subgroups of G/K. Furthermore, this bijection preserves inclusion and index.

- quotient map

Lecture 13

Automorphism

An isomorphism $\varphi : G \to G$. The set of all automorphisms is denoted $Aut(G)$.

- **Identity Automorphism**: $id_G : G \to G$, $g \to g$
- $Aut(G)$: a group under composition
  - $\varphi, \phi \in Aut(G), \varphi\phi \in Aut(G)$ by $\varphi\phi(g) := \varphi(\phi(g))$
  - elements: functions
  - $Aut(\mathbb{Z}_2 \times \mathbb{Z}_2) \cong S_3$ (permuting the three non-zero elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$).
  - For a cyclic group, $|Aut(G)| = \varphi(n) = \text{number of generators}$

Conjugation / Inner Automorphism

Let G be a group and $a \in G$. Conjugation by a is an automorphism of G.

$$\begin{aligned} \varphi_a : G &\to G \\ g &\mapsto aga^{-1} \end{aligned}$$

Such an automorphism is called an **inner automorphism** and the set of all inner automorphisms is denoted $Inn(G)$.

- $Inn(G) < Aut(G)$
- $Inn(G) \lhd Aut(G)$ is a normal subgroup.

Center

Let G be a group. Define the center of G by $Z(G) := \{a \in G : ag = ga \text{ for all } g \in G\}$

- elements that commute with everything

Prop. 53

Let G be a group. Then $G/Z(G) \cong Inn(G)$

Corollary 54

If G is abelian, $Inn(G) \cong \{e\}$

Outer Automorphism Group

$Out(G) := Aut(G)/Inn(G)$

Lecture 14

Let X be a set and define S_X to be the group of permutations of X. That is, the elements are bijections from X to itself, and the operation is composition.

Group Action

Let G be a group and X a set. We say G acts on X, and write $G \curvearrowright X$, if there is a function $G \times X \to X$ such that $e \cdot x = x$ for all $x \in X$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.

- $$\begin{aligned} \psi_g &: X \to X \\ &x \mapsto g \cdot x \text{ for each } g \in G \end{aligned}$$

Prop. 56

Every group action of G on a set X induces a homomorphism $\varphi : G \to S_X$.

Conversely, every homomorphism $\varphi : G \to S_X$ comes from a group action $G \curvearrowright X$.

Cayley's Theorem

Every group is isomorphic to a subgroup of a symmetric group.

- If $|G| = n$, then G is isomorphic to a subgroup of S_n.

Let G act on a set X. For any $x \in X$,

the **orbit** of x is $G\cdot x := \{g\cdot x : g \in G\}$.

- the set of subsets $\{G\cdot x : x \in X\}$ (orbits of all elements) partitions the set X

the **stabiliser** of x is $G_x := \{g \in G : g\cdot x = x\}$

- $G_x < G$

The Orbit-Stabiliser Theorem

Let G be a group acting on a set X. Let $x \in X$.

Then $|G : G_x| = |G\cdot x|$. In particular, if G is finite, then $|G| = |G_x||G\cdot x|$.

Cauchy's Theorem

Let G be a finite group and let p be a prime dividing the size of the group. Then there is an element $g \in G$ such that $|g| = p$.

Corollary 63

Suppose G is an abelian group of order pq for distinct primes p and q. Then G is cyclic.

Let G act on itself by conjugation. For $g \in G$ define the conjugacy class of g, denoted $K(g)$, by the orbit $G\cdot g$. Define the centraliser of g, denoted $Z(g)$ by the stabiliser G_g.

$K(g) := \{aga^{-1} : a \in G\}$

$Z(g) := \{a \in G : ag = ga\}$ $(aga^{-1} = g \to ag = ga)$

fixed set of g

Let G act on a set X. For $g \in G$, define the fixed set of g as $X^g := \{x \in X : g \cdot x = x\}$.

(all elements in X such that $g \cdot x = x$)

The Lemma that is not Burnside's

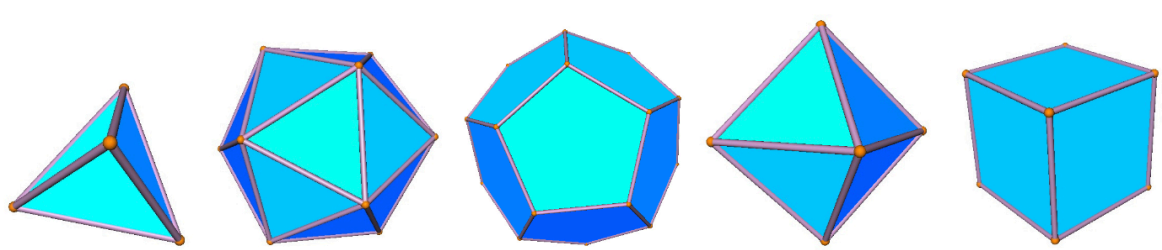
Let G be a finite group acting on a finite set X, and let N be the number of orbits. Then

$$N = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

Lecture 15

Platonic Solids

Platonic Solids	Dual	Rotational symmetry group (isomorphic to)
Tetrahedron	Tetrahedron	A_4
Icosahedron	Dodecahedron	A_5
Dodecahedron	Icosahedron	A_5
Octahedron	Cube	S_4
Cube	Octahedron	S_4



Tetrahedron Icosahedron Dodecahedron Octahedron Cube

Faithful

An action $G \curvearrowright X$ is called faithful if whenever $g \cdot x = x$ for all $x \in X, g = e$.

- Let G act on a set X, and let $\varphi : G \to S_X$ be the corresponding homomorphism. We say the action is faithful if φ is injective.

Generating Set

Let G be a group and H a subgroup. A generating set for H is a subset $S \subset H$ with the property that if any subgroup $K < G$ contains S, then $H < K$. We also say that H is generated by S.

- The alternating group A_n is generated by the set of all 3-cycles.
- Every index 2 subgroup of S_n contains all the 3-cycles. (and the alternating group is the only index 2 subgroup of S_n)

Orthogonal Group

$O(3) = \{A \in GL_3(\mathbb{R}) : AA^T = I\}$

Special Orthogonal Group

$SO(3) = \{A \in O(3) : det(A) = 1\}$

- $|O(3) : SO(3)| = 2$
- Any finite subgroup of $SO(3)$ is isomorphic to one of $A_4, S_4, A_5, \mathbb{Z}_n$, or D_n for $n \geq 2$.

Lecture 16

Prop. 76

Let G be a finite group and $H_1, H_2 \lhd G$ be normal subgroups. If $H_1 \cap H_2 = \{e\}$ and $|H_1||H_2| = |G|$, then $G \cong H_1 \times H_2$.

- Let G be a finite group and let $H_1, ..., H_k$ be a collection of normal subgroups of G. If $H_i \cap \langle H_1, ..., H_{i-1}, H_{i+1}, ..., H_k \rangle = \{e\}$ for all i and $|H_1||H_2| \cdots |H_k| = |G|$, then $G \cong H_1 \times \cdots \times H_k$.

Classification of Finite Abelian Groups

Let G be a finite abelian group. Then $G \cong \mathbb{Z}_{p_1^{n1}} \times \cdots \times \mathbb{Z}_{p_k^{n_k}}$ for some primes $p_1, ..., p_k$ and positive integers $n_1, ..., n_k$. Furthermore, such a presentation is unique up to permuting the factors.

- $\mathbb{Z}_n \times \mathbb{Z}_m \cong \mathbb{Z}_{mn}$ if and only if $gcd(m, n) = 1$
- $\mid G \times H \mid = \mid G \mid \cdot \mid H \mid$

Corollary 79

Let G and H be finite abelian groups. $G \cong H$ if and only if G and H have the same number of elements of order k for all integers $k \geq 1$.